Experiment 9

Perform a t-Test Using a Dataset

Aim: To perform a one-sample t-test in R programming using a dataset imported from a CSV file and interpret the statistical results.

Objectives

1. Import a CSV dataset into R.
2. Understand the concept of a t-test.
3. Perform a one-sample t-test using R.
4. Interpret the t-value and p-value.
5. Make a statistical decision using a 5% significance level.

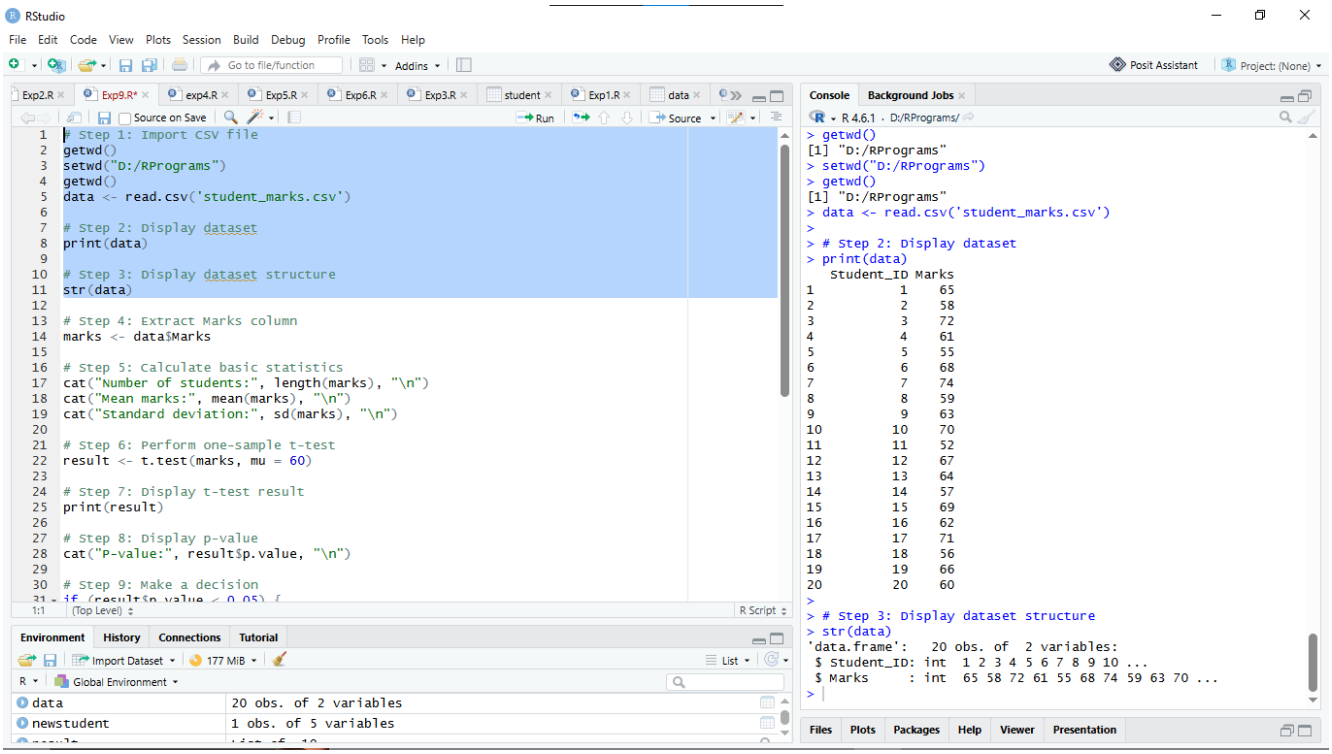
PART B

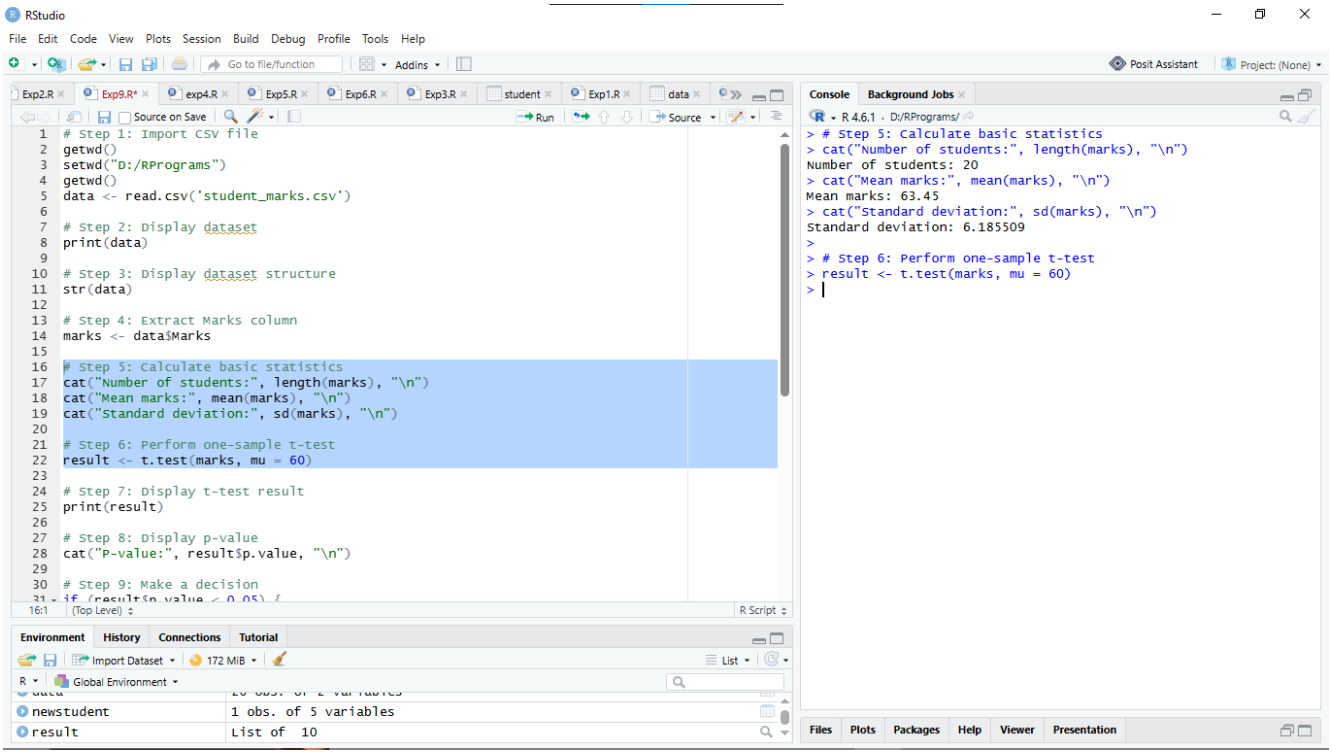
PART B: TO BE COMPLETED BY STUDENTS)

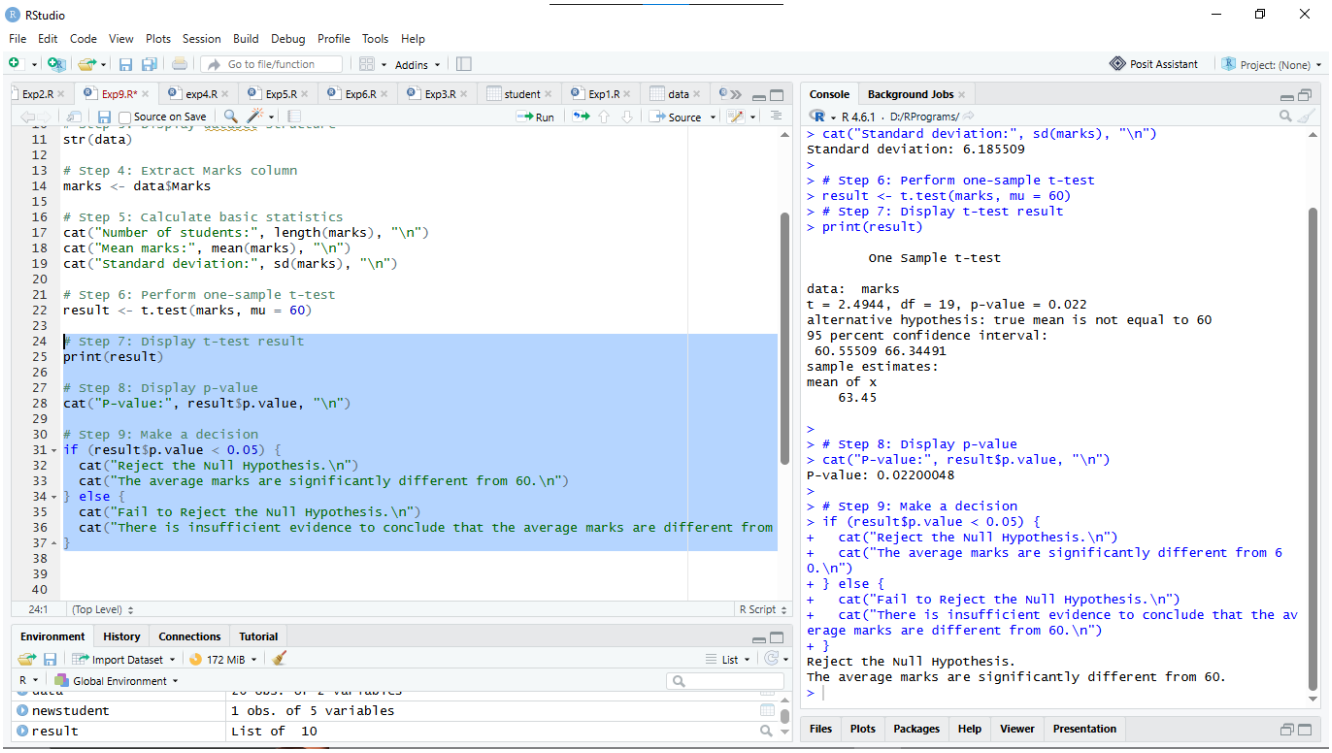
(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

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Class : SE – C	Batch : C2
Date of Experiment:	Date of Submission
Grade:	

Observation : (Paste the screenshots here)




RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Exp2.R Exp9.R* exp4.R Exp5.R Exp6.R Exp3.R student Exp1.R data
Source on Save Run Source
1 # Step 1: Import CSV file
2 getwd()
3 setwd("D:/RPrograms")
4 getwd()
5 data <- read.csv('student_marks.csv')
6
7 # Step 2: Display dataset
8 print(data)
9
10 # Step 3: Display dataset structure
11 str(data)
12
13 # Step 4: Extract Marks column
14 marks <- data$Marks
15
16 # Step 5: Calculate basic statistics
17 cat("Number of students:", length(marks), "\n")
18 cat("Mean marks:", mean(marks), "\n")
19 cat("Standard deviation:", sd(marks), "\n")
20
21 # Step 6: Perform one-sample t-test
22 result <- t.test(marks, mu = 60)
23
24 # Step 7: Display t-test result
25 print(result)
26
27 # Step 8: Display p-value
28 cat("P-value:", result$p.value, "\n")
29
30 # Step 9: Make a decision
16:1 (Top Level) R Script
Environment History Connections Tutorial
Import Dataset 172 MiB List
Global Environment
newstudent 1 obs. of 5 variables
result List of 10
Console Background Jobs
R 4.6.1 · D:/RPrograms/
> # Step 5: Calculate basic statistics
> cat("Number of students:", length(marks), "\n")
Number of students: 20
> cat("Mean marks:", mean(marks), "\n")
Mean marks: 63.45
> cat("Standard deviation:", sd(marks), "\n")
Standard deviation: 6.185509
>
> # Step 6: Perform one-sample t-test
> result <- t.test(marks, mu = 60)
>
Files Plots Packages Help Viewer Presentation


RStudio
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Exp2.R Exp9.R* exp4.R Exp5.R Exp6.R Exp3.R student Exp1.R data
Source on Save Run Source
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24 # Step 7: Display t-test result
25 print(result)
26
27 # Step 8: Display p-value
28 cat("P-value:", result$p.value, "\n")
29
30 # Step 9: Make a decision
31 if (result$p.value < 0.05) {
32 cat("Reject the Null Hypothesis.\n")
33 cat("The average marks are significantly different from 60.\n")
34 } else {
35 cat("Fail to Reject the Null Hypothesis.\n")
36 cat("There is insufficient evidence to conclude that the average marks are different from
37 }
38
39
40
24:1 (Top Level) R Script
Environment History Connections Tutorial
Import Dataset 172 MiB List
Global Environment
newstudent 1 obs. of 5 variables
result List of 10
Console Background Jobs
R 4.6.1 · D:/RPrograms/
> cat("Standard deviation:", sd(marks), "\n")
Standard deviation: 6.185509
>
> # Step 6: Perform one-sample t-test
> result <- t.test(marks, mu = 60)
> # Step 7: Display t-test result
> print(result)

One Sample t-test

data: marks
t = 2.4944, df = 19, p-value = 0.022
alternative hypothesis: true mean is not equal to 60
95 percent confidence interval:
60.55509 66.34491
sample estimates:
mean of x
63.45

>
> # Step 8: Display p-value
> cat("P-value:", result$p.value, "\n")
P-value: 0.02200048
>
> # Step 9: Make a decision
> if (result$p.value < 0.05) {
+ cat("Reject the Null Hypothesis.\n")
+ cat("The average marks are significantly different from 6
0.\n")
+ } else {
+ cat("Fail to Reject the Null Hypothesis.\n")
+ cat("There is insufficient evidence to conclude that the av
erage marks are different from 60.\n")
+ }
Reject the Null Hypothesis.
The average marks are significantly different from 60.
>
Files Plots Packages Help Viewer Presentation

Result:

The output will display the imported dataset, its structure, descriptive statistics, and the t-test result. A typical t-test output contains:

One Sample t-test

data: marks

t = 2.4944

df = 19

p-value = 0.0220

alternative hypothesis: true mean is not equal to 60

95 percent confidence interval: 60.5551 to 66.3449

mean of x: : 63.45

Note: The exact t-value, p-value, confidence interval, and standard deviation should be taken from the R output generated on the student's system.

Conclusion:

If p-value < 0.05: Reject the null hypothesis. There is statistically significant evidence that the average marks are different from 60.

Curiosity Questions

Q1. What is a t-test?

A t-test is a statistical test used to determine whether there is a significant difference between means.

Q2. Why is a t-test used?

A t-test is used to compare sample means and determine whether an observed difference is statistically significant, especially when the population standard deviation is unknown.

Q3. What are the types of t-tests?

The three common types of t-tests are:

1. One-sample t-test
2. Independent two-sample t-test
3. Paired t-test

Q4. Which R function is used for a t-test?

The `t.test()` function is used to perform a t-test in R.

Q5. What is a one-sample t-test?

A one-sample t-test compares the mean of a single sample with a specified or hypothesized population mean.

Q6. What is the null hypothesis?

The null hypothesis (H_0) is a statement that assumes there is no significant difference or effect. In this experiment, H_0: $\mu = 60$.

Q7. What is the alternative hypothesis?

The alternative hypothesis (H_1) states that the population mean is different from the hypothesized value. In this experiment, H_1: $\mu \neq 60$.

Q8. What is the significance level?

The significance level (α) is the threshold used to decide whether to reject the null hypothesis. In this experiment, $\alpha = 0.05$ or 5%.

Q9. What is a p-value?

A p-value is the probability, assuming the null hypothesis is true, of obtaining a test statistic at least as extreme as the observed one.

Q10. When do we reject the null hypothesis?

We reject the null hypothesis when the p-value is less than the significance level, usually 0.05.

Q11. What does `mu = 60` mean in `t.test()`?

It specifies the hypothesized population mean as 60 marks.

Q12. What is a t-statistic?

The t-statistic measures how far the sample mean is from the hypothesized population mean relative to the estimated standard error.

Q13. What are degrees of freedom?

Degrees of freedom represent the number of independent values available for estimating statistical quantities.

Q14. How are degrees of freedom calculated for a one-sample t-test?

The formula is:

$$ df = n - 1 $$

For 20 students, $df = 20 - 1 = 19$.

Q15. How do you import a CSV file in R?

The `read.csv()` function is used to import a CSV file.

```
data <- read.csv("student_marks.csv")
```

Q16. How do you calculate the mean in R?

The `mean()` function calculates the arithmetic mean.

```
mean(marks)
```

Q17. How do you calculate standard deviation in R?

The `sd()` function calculates the sample standard deviation.

```
sd(marks)
```

Q18. What is a two-tailed t-test?

A two-tailed t-test checks whether a population mean is significantly different from a specified value in either direction (greater or smaller).

Q19. What is a paired t-test?

A paired t-test compares two related measurements, such as students' marks before and after training.

Q20. What is an independent two-sample t-test?

An independent two-sample t-test compares the means of two separate, independent groups.

Q21. What are the assumptions of a t-test?

The main assumptions are:

1. Observations are independent.
2. Data are approximately normally distributed, particularly for small samples.
3. The data are numerical.
4. For the standard pooled two-sample t-test, the two groups have equal population variances.

Q22. What is a confidence interval?

A confidence interval is a range of values used to estimate an unknown population parameter at a specified confidence level.

Q23. What is the default confidence level in `t.test()`?

The default confidence level is 95%.

Q24. How do you perform a one-tailed t-test in R?

Use the `alternative` argument.

For testing whether the mean is greater than 60:

```
t.test(marks, mu = 60, alternative = "greater")
```

For testing whether the mean is less than 60:

```
t.test(marks, mu = 60, alternative = "less")
```

Q25. What is the conclusion of a t-test based on the p-value?

If the p-value is less than 0.05, reject the null hypothesis. If the p-value is greater than or equal to 0.05, fail to reject the null hypothesis.

1. What is the conclusion of a t-test based on the p-value?